DFD6450

Fully Automatic Dicing Saw

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Fully automatic dicing saw with parallel dual spindles

Φ200 mm Parallel dual spindle Package Singulation

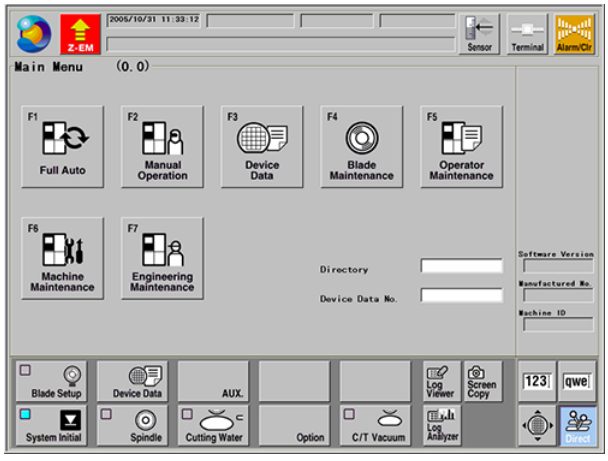


A parallel dual design unique to DISCO

Featuring parallel dual-spindles, DFD6450 inherits the basic concept of the existing model, DFD651, and has been added to the DFD6000 series lineup with superior usability and processing quality. Designed to handle a wide range of dicing and grooving applications, the DFD6450 processes silicon, compound semiconductor wafers, ceramics and other materials used for electronic component applications. In addition, the DFD6450 also processes glass and other materials used for optical component applications.The DFD6450's parallel dual spindle design is unique to DISCO and offers several benefits. As the spindles are supported from the rear of the machine, they are easily accessible from the front for blade changes and other maintenance work. The design also supports specialized applications such as mounting multiple blades on each spindle for package singulation.

Improved usability

The 15-inch LCD touch panel with GUI (Graphical User Interface) allows for easy and intuitive operation. In addition, a spindle shaft-lock feature that automatically secures the spindle axis has been installed to make blade replacement quick and easy. DFD6450 also has a cutting water flow rate control feature, making it possible to set the cutting water flow rate for each set of device data.



Control screen

Advanced options

The equipment can have a transformer, UPS, CO_2 injector, or booster pump unit built-in by revising the equipment layout while maintaining an equipment footprint equal to that of the existing model. In addition, an atomizing nozzle mechanism for the spinner section with high cleaning effect on workpieces and ionizer specifications for preventing electrostatic build up during workpiece transfer can be selected.

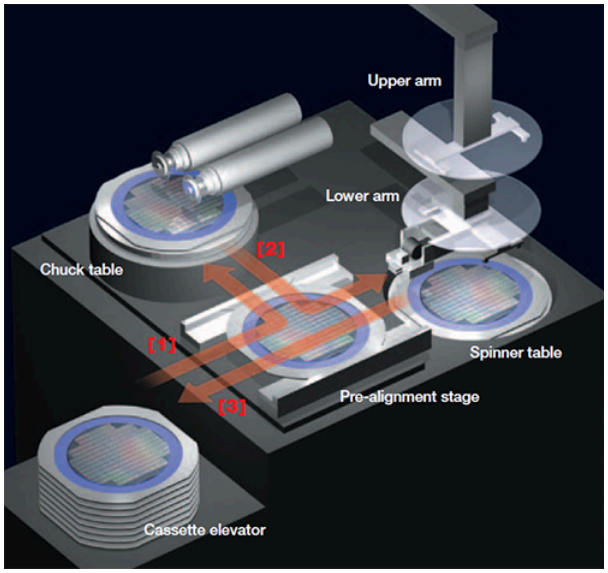
Spindle lineup

A range of spindles supports diverse workpieces and applications.

- **[Standard]**
  1.0 kW air spindle
  (uses 2" blades, rated torque: 0.16 N·m).
- **[Options]**
  1.0 kW air spindle with higher torque
  (uses 2" blades, rated torque: 0.24 N·m)
  2.2 kW air spindle
  (uses 3" blades, rated torque: 0.7N·m)

DFD6450 Operation flow

1. Lower arm moves the workpiece from the cassette to the pre-alignment stage. Lower arm moves the workpiece to the chuck table → **cutting** →
2. Upper arm moves the workpiece to the spinner table → **cleaning and drying** →
3. Lower arm returns the workpiece to the cassette



Specifications

Specification		Unit	1.0 kW	2.2 kW
Max. workpiece size		-	Φ8 inch	
X-axis	Cutting range	mm	250	
	Cutting speed	mm/s	0.1 ~ 600	
Y1-axis	Cutting range	mm	250	
	Index step	mm	0.0001	
	Positioning accuracy	mm	Within 0.003/250 (Single error) Within 0.002/5	
Y2-axis	Cutting range	mm	30	
	Index step	mm	0.0001	
	Positioning accuracy	mm	Within 0.002/30	
Z-axis	Max. stroke	mm	35.2	
	Moving resolution	mm	0.00005	
	Repeatability accuracy	mm	0.001	
θ-axis	Max. rotating angle	deg	380	
Spindle	Rated output	kW	1.0	2.2
	Rated torque	N・m	0.16, 0.24 (higher torque)	0.7
	Rotation speed range	min^{-1}	6,000 ~ 60,000、 3,000 ~ 40,000 (higher torque)	3,000 ~ 30,000
Equipment dimensions(W×D×H)		mm	1,120 x 1,500 x 1,600	
Equipment weight		kg	Approx. 1,400	

*Product appearance, features, specifications, and other details may change due to technical modifications.
*Please read the standard specification sheet thoroughly before use.

Product Lineup







	DFD6240	DFD6340	DFD6341	DFD6342	DFD6361
Machine type	Fully Automatic	Fully Automatic	Fully Automatic	Fully Automatic	Fully Automatic
Spindle	1	2	2	2	2
C/T	1	1	1	1	1
Machine dimentions (W × D × H)	900 × 1190 × 1800	1180 × 1110 × 1850	1180 × 1080 × 1820	1180 × 1080 × 1800	1200 × 1550 × 1800 1
Machine weight (kg)	1200	1600	1500	1240	2050

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